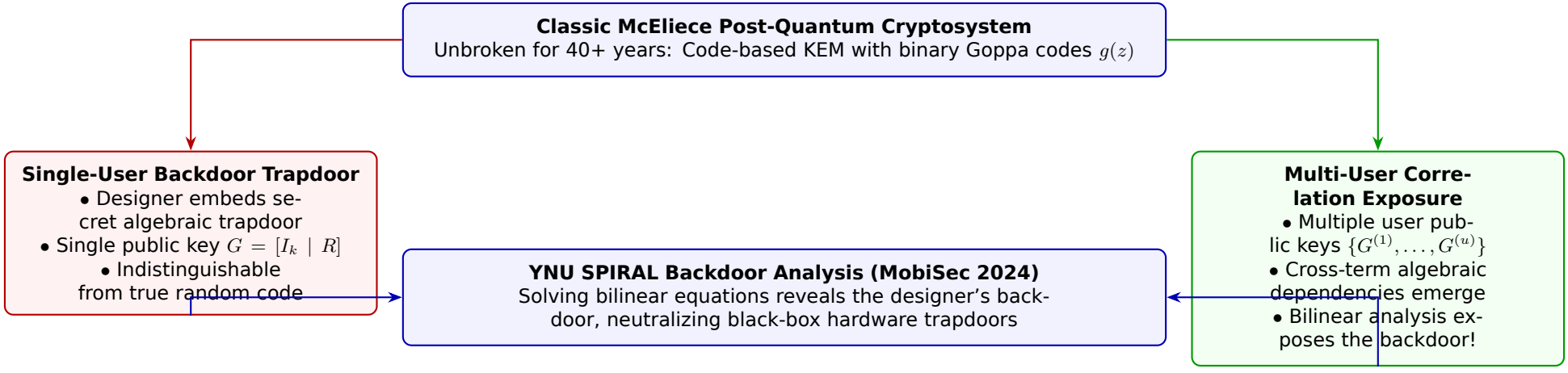

Classic McEliece Post-Quantum Cryptosystem
Unbroken for 40+ years: Code-based KEM with binary Goppa codes $g(z)$
Single-User Backdoor Trapdoor
• Designer embeds secret algebraic trapdoor
• Single public key $G = [I_k \mid R]$
• Indistinguishable from true random code
YNU SPIRAL Backdoor Analysis (MobiSec 2024)
Solving bilinear equations reveals the designer's backdoor, neutralizing black-box hardware trapdoors
Multi-User Correlation Exposure
• Multiple user public keys $\{G^{(1)}, \ldots, G^{(u)}\}$
• Cross-term algebraic dependencies emerge
• Bilinear analysis exposes the backdoor!